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Identifying New Clotting Factor XIa Inhibitors in Virtual High-Throughput Screens Using PCA-GA-SVM Models and Signature

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Blood Clotting Factor XI is an important actor in the clotting mechanism: it activates downstream zymogen involved in the clotting process. It can be targeted for activation or inhibition depending on treatment goals to enhance or inhibit clotting. In terms of antithrombosis treatment, Factor XI has emerged as a promising target to focus on. In this work, an iterative virtual high-throughput screening pipeline was proposed that can supplement current efforts to find inhibitors. The first iteration identified 11 compounds to test with 3 active for a hit-rate of 27.3%. The second iteration of the pipeline identified another 11 compounds to test with 7 active for a hit-rate of 63.6%. © 2018 American Institute of Chemical Engineers *Biotechnol. Prog.*, 34:1553–1565, 2018
Keywords: Factor XIa, virtual high-throughput screening, data mining, QSAR, drug discovery, Signature

Introduction

Factor XI is a clotting factor whose absence was noticed in patients with mild to moderate hemophilia.[1,2] A 160 kDa homodimer, made of two 80 kDa monomers,[3] Factor XI is comprised of 607 amino acid residues[3] and is encoded by the 23 Kb long FXI gene, with 15 exons and 15 introns,[4] found in the 4th chromosome.[5] The 160 kDa homodimer is produced by the liver and circulates the body as an inactive zymogen at a concentration of about 4 μg/mL.[6]

Classical models of hemostasis propose a cascade reaction in which precursors activate and amplify downstream signals to stop blood loss. In this model, activated Factor XII binds to Factor XI, triggering a Factor XI conformation change to become activated Factor XI (Factor XIa).[7] Factor XIa then activates other factors that are active in the clotting process. However, the classical model conflicts with other case studies in which patients with low precursor levels exhibit no symptoms of hemophilia while patients with low Factor XI levels do.[4] Later, work determined Factor XI can also be activated by the downstream product thrombin in conjunction with other Factor XIa molecules and negatively charged surfaces.[8,9] This reduces the need for precursors while also increasing the role of Factor XI as a clotting-signal amplifier to convert more prothrombin to thrombin thereby accelerating the conversion of fibrinogen to fibrin.[9]

While a symptom of Factor XI deficiency is hemophilia, there is no strict correlation between the two and treatment is on a case-by-case basis.[10,11] Conversely, those people with elevated levels of Factor XI may be at risk of clotting too much, resulting in thrombosis.[12–15] In cases of thrombosis, inhibiting the proclotting tendencies of Factor XI may be a desired course of treatment. Researchers testing the antithrombic effects of deficient Factor XII levels have also tested deficient levels of Factor XI, reasoning that similar effects should be observed as Factor XI are downstream of Factor XII and, therefore, more directly involved with the formation of clots.[16,17] Testing results tend to support this hypothesis, with decreased thrombosis stability in mice genetically modified for decreased Factor XI levels.[16,17] In light of these results, Factor XIa becomes a desirable inhibition target for antithrombosis treatment.

Researchers have been targeting Factor XI for inhibition since the 1970s, starting with four protease inhibitors: α1-protease inhibitor,[18] antithrombin III,[19] C1-inhibitor,[20] and α2-plasmid inhibitor.[21] Since then, derivatives of the original protease inhibitors[22] and other kinds of inhibitors[23–29] have been developed. To maximize the likelihood of developing a safe antithrombosis treatment, other techniques should be utilized to supplement traditional discovery methods. The introduction of computers and subsequent exponential growth in computational power has made it possible to design new molecules or find existing ones that may be drug candidates.[30] In this work, we harness this increased computational power to perform a virtual high-throughput screen (vHTS) to find potential new inhibitors for Factor XIa.

Consider the number of organic molecules that are the possible drug candidates. With the different dimensions involved

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(e.g., kinds of atoms, number of atoms, kinds of bonds, number of bonds, branching, cyclization, etc.), the number of organic molecules grows exponentially. For example, a conservative estimate of the number of molecules that can be made from combining 30 C, N, O, and S atoms is 10^{60},[31] including both theoretical (yet to be synthesized/discovered) and actual(already synthesized/discovered) molecules. As Bohacek reasoned,[31] this number "grows steeply" with molecular weight and, for the purposes of illustration, can serve as an estimate of the drug candidate pool size. High-throughput screening (HTS), developed with the goals of minimizing cost and increasing speed by reducing the reagents needed and testing more candidates in parallel,[32–34] can systematically explore parts of the candidate pool in the form of compound libraries. Unfortunately, HTS still spends almost all resources testing inactive candidates and thus has low "hit-rates."[35] Experts can preprocess compound libraries, known as focusing, by removing candidates from the testing phase that are likely inactive, however the effect is limited.[35] Fortunately, a result of HTS experiments is large amounts of experimental data. By cross-referencing this with other available characterization and experimental data (e.g., PubChem,[36,37] CHEMBL,[38] and ZINC[39]) from "big data,"[40] new and alternative ways of using this data can be developed. Characterization and experimental data can be used to develop mathematical models for the purposes of *virtually* focusing a compound library in a vHTS. In this way, virtual focusing can be used in conjunction with expert knowledge to focus compound libraries. Thus, additional information gleaned from previous efforts can guide future work and increase the likelihood of successfully identifying new/novel inhibitors.

The two major categories of vHTS approaches are molecular simulation (e.g., AutoDock, DOCK, Flex, AMBER, GROMACS, CHARMM),[41–52] and ligand-based scoring.[49,53–55] Molecular simulation uses models and concepts from physics, chemistry, and mathematics to make optimized ligand/substrate interaction predictions. Although molecular simulations require minimal experimental data, it is computationally expensive and requires numerical solution convergence for confidence. Ligand-based scoring uses known ligand data to find other possible ligands. Ligand-based scoring is computationally cheaper, but requires previous experimental data to make predictions and is often limited to molecules similar to known ligands as prediction power drops with extrapolation. Hybrid/mixed approaches can be used, but the strengths and deficiencies of each still exist.[56]

The approach used to identify Factor XIa inhibitors was previously introduced in our lab[57,58] and is available for reference in Figure 1. The approach attempts to maximize data usage by combining three ligand-based vHTS methods: (i) classification as active/inactive, (ii) prediction of activity based on a quantitative structure–activity relationship (QSAR) created from the regression of the available data, and (iii) similarity to known active ligands. The predictive models are created from experimental and chemical structure data and used to screen compound databases/libraries. Developed with scaling in mind, the approach screens, all of PubChem's Compound database (currently about 72 million compounds) or even larger ones to make a manageable focused compound library. It should be noted that screening PubChem Compound for hits limits consideration to compounds that are thermodynamically stable enough to be synthesized only. This is essential to validating model predictions via experimentation as theoretical compounds cannot be tested for biological activity.

The vHTS models are a combination of three different algorithms and methods: principal component analysis (PCA), support vector machine (SVM), and genetic algorithm (GA). The way these three algorithms/methods are combined is described in the *Experimental Section.* Signature, a molecular structure fragmentation[59,60] and elucidation[61] technique, has a history of classification and QSAR work including substrate/receptor selection,[26,27,57,62] protein/protein selection[63] and molecular design,[64–69] is the foundation of the approach and is used to create the inputs into our algorithm from the molecules in the data set. Molecule fragmentation using Signature is demonstrated in Figure 2.

This work is part of a more extensive study of the previously presented pipeline[57,58] to determine its effectiveness on different protein/ligand systems and the effect of data set size and active/inactive ligand distribution on predictive power. Previously, the pipeline has been applied to identify inhibitors of Cathepsin L[57] and Factor XIIa.[58] For this work, the pipeline used available Factor XIa ligand structure and activity data from PubChem Bioassay data set AID 846[70] (Figure 1; step 1). Once the predictive classification and QSAR models were trained, the entire PubChem Compound database (72 million) was screened using the models (Figure 1; step 2) for synthesized compounds with potential biological activity. Identified leads (Figure 1; step 3) were experimentally tested (Figure 1; step 4) to (i) identify new/novel Factor XIa inhibitors, (ii) validate predictions made, and (iii) evaluate the models and the pipeline that created them. The experimental protocol used is based on the protocol from the depositors of AID 846, scaled for use with 96-well plates instead of 384-well plates.[70] After experimental validation, the models are retrained using the original training data augmented with the newly obtained experimental data. A second vHTS was conducted with another experimental validation to determine if any additional inhibitors can be identified as well as to determine if the retrained models performed better than the previous ones.

Materials and Methods

Creating/training predictive models

Once known receptor-ligand data have been found and compiled into a training set, the compounds are deconstructed into atomic Signatures as shown in Figure 2. Based on these atomic Signatures, models are created and trained. The model creation and training process is comprised of PCA, GA, and SVM. PCA is generally used to create linearly independent variables (principal components) that capture the variance in the data via combining the weighed original variables.[71] However, the new principal components often lose most of the physical significance of the original variables. To avoid this loss in physical meaning, our work examines the weights associated with each atomic Signatures in creating the principal components and selects for those that contribute the most (i.e., the most weight) when making the principal components. Using PCA in this way allows us to identify atomic Signatures that are important to explaining data variance while retaining the physical meaning of each atomic Signature. The PCA-identified significant atomic Signatures are used to create and train the vHTS models.

The vHTS models are created through the interactions of two different algorithms: GA and SVM. GA operates by selecting for combinations of genes that maximize or

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minimize a desired property/parameter.[72] In this work, the genes are the atomic Signatures. As predictive power is the desired property of these models, prediction accuracy is the optimized parameter. To create the models, each combination of genes created by GA is sent to SVM to create predictive models.

SVM models depend on the kind of information it is given during training: classification data (e.g., 1/+1) will yield a classification model while continuous data (e.g., IC_{50}) will yield models predicting continuous values of the kind with which it was trained. Cross-validation is a data-handling approach that repeatedly divides the available data into either the training or testing groups and using the data to train or evaluate the models, respectively. Cross-validation accuracy, or the accuracy of the models when applied to the testing group of data, can also be interpreted as a measure of a model's predictive ability. Thus cross-validation accuracy is the optimized variable and is reported back to GA.

With the cross-validation accuracy of different combinations of genes, GA selects for the gene combinations with the highest prediction accuracy to create new combinations to test. The new combinations are created through well-defined genetic operators. New SVM models are created with the new gene combinations and more prediction accuracy is reported back. The GA-SVM interaction is repeated until either the parameter reaches a desired value or has not been improved upon over a given number of repeats.

As with any modeling work, one must always be wary of overfitting. Overfitting occurs when the models not only capture and describe the correlation between the independent and dependent variables but also the noise. To limit any overfitting, we took advantage of the features of several different parts of our approach. First is the use of PCA: PCA removes variables that do not correlate with activity and are possible sources of overfitting. Second is the use of GA: the stochastic, perturbational nature of recombination to create new individuals in a generation, as well as the testing of the best individuals in a generation under different conditions, results in models that are robust and less likely to be overfit rather than specialized models that are more likely to be overfit. Last is the use of cross-validation: a portion of the data is sequestered from the training process as the testing set and overfitted models are not likely to perform well when applied to the testing set. Together, the features of different components in the approach attempt to limit any overfitting that may occur.

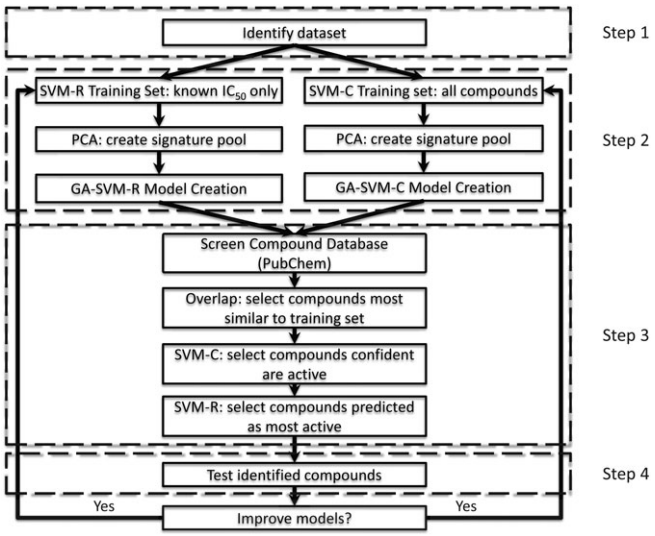


Figure 1. The approach has four main steps: (1) identify a target data set, (2) training predictive classification and QSAR models using the identified data set, (3) screen a compound library with the classification and QSAR models, and (4) experimentally validate model predictions. Signature is used to process molecules from the data set or libraries into inputs for our approach. Adapted from Chen, J and Visco, D.P. Eur J Med Chem, 2017, 140, 31–41. Copyright © 2017 Elsevier Masson SAS. All rights reserved.

Screening with predictive models

Once models have been trained, they are used to screen compound databases (here, PubChem Compound) to identify candidates. The models are evaluated using two different metrics, the first during training to select the best vHTS models and the second after experimental validation. The first, accuracy, is defined by Weis[26] as:

$$\text{Accuracy} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}},$$

where "TP" and "TN" mean true positive and true negative, respectively and "FP" and "FN" mean false positive and false negative, respectively. In our work, the accuracy used was cross-validation accuracy and training accuracy, the latter as a tie-breaker if models had the same cross-validation accuracy but differing training accuracy. Precision, the metric used after experimental validation as an a posteriori metric of predictive power, is defined by Weis[26] as:

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}},$$

where "TP" and "FP" mean true positive and false positive. Hit-rate, defined in the same way, is the term more commonly used in HTS.

In modeling, there is an inverse relationship between power and extrapolation: model prediction power decreases as extrapolation increases. Similar previous work has verified this relationship when considering molecules with atomic Signatures the vHTS models are not trained on.[26] In the many ways of characterizing similarity,[73] we use a metric called overlap that was developed based on the set-theoretic definition of the Tanimoto Coefficient[73] and defined by Weis[26] as:

$$\Omega = \frac{x_{[\min,\max]}}{x_{\text{total}}},$$

where $x_{[\min,\ \max]}$ is the total number of unique atomic signatures in the compound that falls within the maximum and minimum occurrences observed in the training set.

Model generation parameters

All modeling processes were done on dual Intel Xeon processors (E5-2697W, 3.10 GHz, 48 independent threads). The 48 independent threads were utilized to run 48 iterations of model generations with different initial conditions. Additionally, 32 of the 48 independent threads were used to split the screening load by splitting the PubChem Compound database (72 million compounds) into 32 subsets and each subset

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screened on its own thread. PCA, GA, and SVM were done using R Statistical Software: PCA using the "eigen" function, GA using the "ga" function in the "GA" package,[74] SVM using the "ksvm" function in the "kernlab" package.[75] Parameters for GA were as follows: elitism rate = 0.7, crossover rate = 0.8, mutation rate = 0.1, population size = 1,000, maximum iterations = 1,000, stop after 100 iterations of no improvement. Parameters for SVM were as follows: cost ranges from 0.01 to 1 with step size 0.01, 10-fold cross-validation, ν = 0.2, linear kernel. ROC curves for model evaluation were created using the" ROCR" package.[76]

Fluorescent factor XIa inhibitor screening assay

Assay materials. Assay buffer: H_2O with 50 mM Tris, 150 mM NaCl, 0.02% Tween 20, and adjusted to pH 7.4. Enzyme Solution: Factor XIa (final concentration 0.23 μg/mL). Substrate Solution: Boc-Glu-Ala-Arg-AMC (final concentration 15 μM). Plate: Corning black polystyrene 96-well, flat bottom. Trizma Base, Tween 20, and Corning flat-bottom, black polystyrene 96-well plates are purchased from Sigma Aldrich (St. Louis, MO), Factor XIIa from Enzyme Research Laboratories (Cat ID HFXIa 1111a; South Bend, IN), and the Boc-Glu-Ala-Arg-AMC from Bachem (Cat ID I-1575.0050, Rorrance, CA). All testing compounds were purchased via Molport (Riga, Latvia).

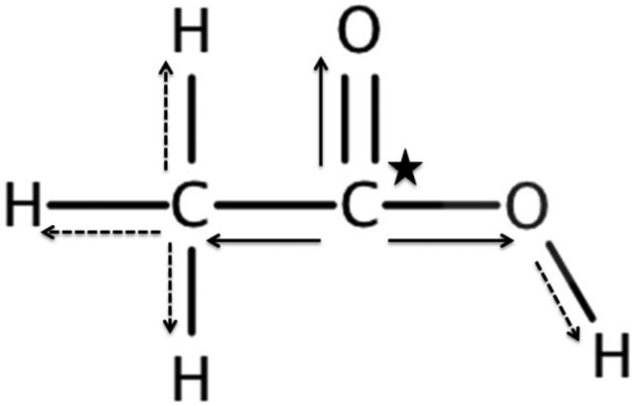


Figure 2. Molecular structure transformation into Signature fragments. Starting from the root atom, like the starred carbon, atomic neighbors, and connections, without backtracking, are noted to a predetermined distance (height) away. When height = 0, only the root atom is noted. When height = 1 (solid arrows), the primary atomic neighbors and their bonds to the root atom are noted. When height = 2 (dashed arrow), the notation from height = 1 is amended to include the root atom's secondary atomic neighbors and their connecting bonds to the primary neighbors. The record for a single root atom is known as an atomic Signature; the collection of atomic Signatures for all atoms in the molecule is known as the molecular Signature. Reproduced from Chen, J and Visco, D.P. Eur J Med Chem, 2017, 140, 31–41. Copyright © 2017 Elsevier Masson SAS. All rights reserved.

Assay protocol

1. Serial dilute identified compounds at 50× concentration in DMSO: 8 fourfold dilutions from 2.5 mM to 152.6 nM. Final testing concentration from 50μM to 3.05 nM.
2. Fill 96-well plate with 50 μL substrate solution.
3. Add 2 μL of compound from step 1 to corresponding wells in the plate.
4. Add 50 μL enzyme to wells with compound added.
5. Control: Add 50 μL enzyme to rows A–D of columns 9–12.
6. Blank: Add 50 μL buffer to rows E–H of columns 9–12.
7. Incubate for 1 h at room temperature.
8. Read fluorescence (excitation 355, emission 460) on Tecan M200.

Percent Inhibition determination. Percent inhibition was calculated using the following relationship:

$$\%\text{inhibition} = \left(1 - \frac{\text{signal-}\overline{\text{blank}}}{\text{control-}\overline{\text{blank}}} \times 100\right).$$

To calculate the IC_{50} value, a linear interpolation of the closest data point above and below 50% inhibition was conducted. If 50% inhibition was not included in the range of inhibition values, the compound was deemed inactive. There were no compounds for which the IC_{50} value was below the minimum testing concentration.

Results and Discussion

First pass classification and QSAR model creation, vHTS, and validation results

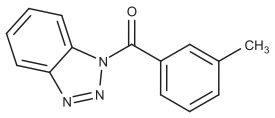
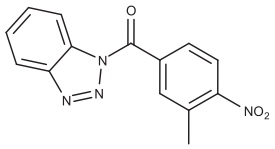
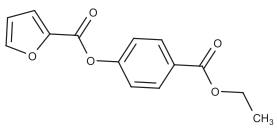
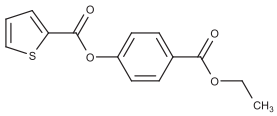
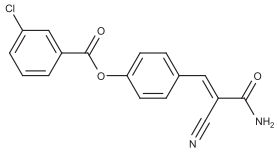
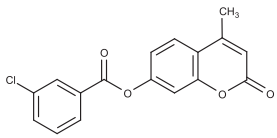
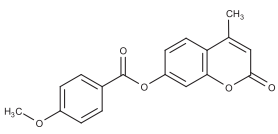
AID 846 contains 91 actives with known corresponding IC_{50} values and 198 inactives.[70] The active/inactive classes provide the necessary information to train the classification models (SVM-C), while the known IC_{50} values of the actives provides necessary information to train the QSAR regression models (SVM-R). The highest concentration tested in AID 846 was 50 μM. As the ceiling of known IC_{50} values, it is also the ceiling of acceptable IC_{50} predictions. Accepting any predictions above 50 μM would be extrapolating the models. As previously mentioned, there is an inverse relationship between extrapolation and predictive power. As accurately predicting compound activity was the goal, this limitation was implemented to prevent extrapolation. The molecular structure of the compounds in AID 846 was transformed into atomic Signatures of heights 0, 1, and 2, 1,798 atomic Signatures in total. PCA was used to filter for significant atomic Signatures and only these were used in the construction of the predictive classification and QSAR models using the GA and SVM algorithms. It is important to note the classification models are created using all data available, as all compounds were assigned a class, whereas the QSAR models are created from a subset of data containing only the active compounds, as only active compounds possess known IC_{50} values. Similarity was used as the initial filter when processing the results of vHTS

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Table 1. First Pass Model Summary and Statistics

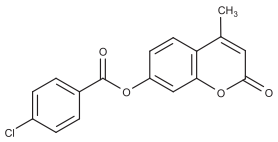
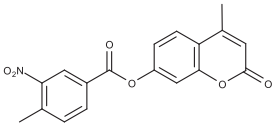
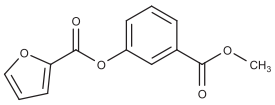
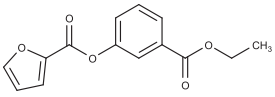
	SVM-C	SVM-R
Training set composition	289 compounds (91 active, 198 inactive)	91 compounds active w/IC_{50}
Training set Signatures	13 h = 0; 200 h = 1; 1585 h = 2; 1,798 total	10 h = 0; 117 h = 1; 552 h = 2; 679 total
PCA results	168 of 1,798 atomic Signatures	262 of 679 atomic Signatures
Models created	1	1
Training error	0.06	0.50
Cross-validation Error	0.06	0.55

Table 2. vHTS First-Pass Validation Results

Structure	CID	Predicted IC_{50}[μM]	Experimental IC_{50}[μM]
	834536	1.05	19.4
	764214	0.91	43.7
	17178134	2.67	37.2
	4026876	2.13	>50
	6042862	6.14	>50
	682556	2.71	>50
	812347	2.72	>50

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Table 2. Continued

Structure	CID	Predicted IC_{50}[μM]	Experimental IC_{50}[μM]
	802465	2.71	>50
	733816	2.72	>50
	17178138	4.77	>50
	17131127	5.48	>50

Compounds selected are commercially available, economically viable and passed the following criteria: predicted IC_{50} < 50μM, SVM-C scores >2 and overlap = 1. Compounds were tested in triplicate. Compounds with reported IC_{50} values are active across the triplicates. The reported experimental IC_{50} value is the mean of the triplicates. CID is the compound's PubChem ID number.

using the metric "overlap,"[26] described in the *Experimental Section.* Briefly, the overlap is the extension of similarity metrics that are used for molecule-to-molecule comparisons by determining if all atomic Signatures in the molecule are present in the training set. To ensure the highest accuracy in model prediction, the overlap is with respect to the QSAR training set as all compounds in the QSAR training set are in the classification training set but not the inverse. A summary of the modeling results is provided in Table 1.

The SVM-C and SVM-R models' predictive abilities were evaluated using receiver operating characteristic (ROC) curves, shown in Figure 3. The shape of the ROC for the SVM-C model indicates there is a threshold value that mostly separates the classes while the area under curve (AUC) indicates the class predictions by the model are a function of model abilities rather than random chance. It is noted that the SVM-R is constructed using a subset of data while the ROC curve was done using all available data. The SVM-R model should perform better than the ROC curve suggests if extrapolation to atomic Signatures not used during training is limited or eliminated.

Both the classification and QSAR models detailed in Table 1 were applied to screen the PubChem Compound database (about 72 million). The following criteria were used to filter the results of the vHTS to create a focused database:

1. Overlap must equal 1 (i.e., all atomic Signatures in the candidate molecule must be found in the active subset of the training set)
2. SVM-C score must be greater than 2
3. Predicted IC_{50} must be less than 50 μM

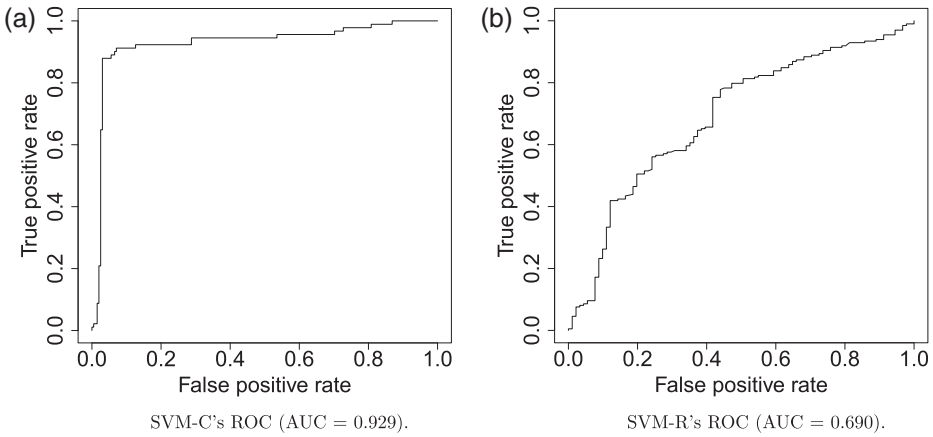


Figure 3. The ROC curve for the SVM-C and SVM-R models. The curves indicate they both contribute to correctly predicting compound activity: the SVM-C curve approaches (1,1) indicating there is a threshold that mostly divides the actives compounds from inactive compounds while the AUC of both curves show active predictions are likely the result of model ability rather than random chance.

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Table 3. Second Pass Model Summary and Statistics

	SVM-C	SVM-R
Training set composition	300 compounds (94 active, 206 inactive)	94 compounds active w/IC_{50}
Training set Signatures	13 h = 0; 200 h = 1; 1,585 h = 2; 1,798 total	10 h = 0; 117 h = 1; 552 h = 2; 679 total
PCA results	172 of 1,798 atomic Signatures	232 of 679 atomic Signatures
Models created	1	1
Training error	0.08	0.61
Cross-validation Error	0.09	0.54

The overlap = 1 is with respect to the active subset of the training set and the IC_{50} ceiling criteria was used because predictive power decreases rapidly with increasing extrapolation: all known IC_{50} were below 50μM, while the SVM-R models were only trained on the compounds with known IC_{50} values. SVM-C score threshold was set at 2 instead of 0 to increase the confidence and likelihood compounds are correctly classified as active. One hundred ten compounds met the specified criteria, but only 11 were commercially available and economically viable with low cost-per-unit. These compounds, along with 2 active and 2 inactive controls, were purchased. Experimental validation confirmed 3 of the 11 compounds were active for a hit-rate of 27.3%. The full experimental validation results are detailed in Table 2.

Second pass classification and QSAR model creation, vHTS, and validation results

The 27.3% hit-rate of the first pass was higher than typically expected from HTS.[35] However, our previous work has demonstrated increased predictive performance in the second pass following retraining with the inclusion of validation data to the original data set. Thus, the AID 846 data set was supplemented with the data detailed in Table 2 and the models retrained. The resulting data set now has 300 compounds total: 94 actives with known IC_{50} and 206 inactive. Following the same protocol for generating models from the first pass, the retrained models were created and detailed in Table 3.

SVM-C and SVM-R model prediction ability was again evaluated using ROC curves, shown in Figure 4. Again, the curves suggest there is a benefit from using the models: the shape of the SVM-C ROC curve indicates, there is a threshold that mostly separates the active and inactive compounds while the AUC values indicate predictions is due to model ability rather than random chance. It is interesting to note the drop in performance according to the ROC curve but rise in hit-rate for the second round validation experiments. Our explanation for this change between the first and second pass is mentioned in more detail in the *Model Discussion.*

After vHTS and applying the previous criteria, too few compounds were economically viable. Thus, the second criteria regarding the classification score was relaxed from 2 to 1.5 to allow more candidates to be examined. After relaxing the SVM-C criterion, 116 compounds met the new criteria but with 11 economically viable. Experimental validation found 7 of the 11 compounds to be active for a hit-rate of 63.6%. It should be noted that 2 of the 11 were also found as likely Factor XIIa inhibitors in our previous work.[58] The results of the validation are detailed in Table 4.

One of the compounds from Table 4, CID 5440853, was similar to a known pan-assay interference compound (PAINS),[77] specifically a compound identified as an aggregate by Ferreira.[78] The "PAINS" filter from ZINC15[39] was used to determine if the compound in question was similar to the one

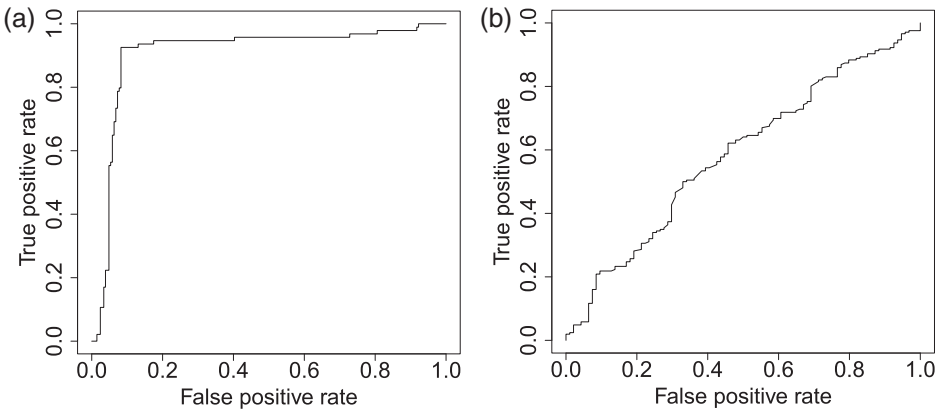
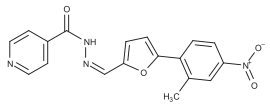
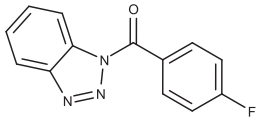
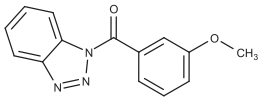
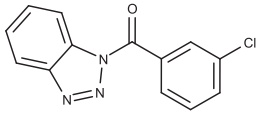
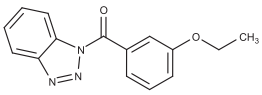
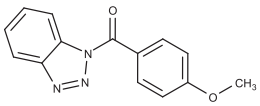
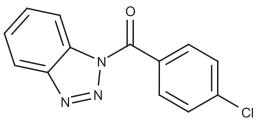
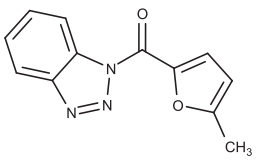
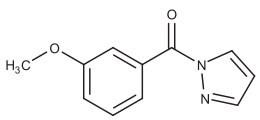
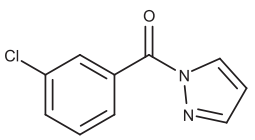


Figure 4. ROC curves of the retrained SVM-C and SVM-R models. The ROC curves indicate there is at least a slight benefit from using the models over random chance. SVM-C's ROC approaching (1,1) suggest there is a threshold that mostly separates the active compounds from the inactive compounds while the AUC suggest it is due to model ability rather than random chance. The SVM-R ROC curve is slightly better than random chance but this should be the baseline performance. SVM-R performance should rise as extrapolation is limited or removed.

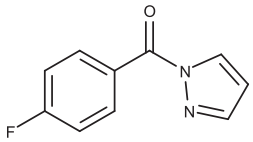
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Table 4. vHTS Second Pass Validation Results

Structure	CID	Predicted IC_{50}[μM]	Experimental IC_{50}[μM]
	5440853	7.93	29.1
	785793	4.13	30.3
	5146207	4.06	32.4
	743608	2.25	15.0
	4957387	4.67	19.7
	710644*	2.05	0.58
	746939*	2.25	>50
	5166982	1.55	0.73
	5101792	2.66	>50
	4992436	1.47	>50

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Table 4. Continued

Structure	CID	Predicted IC_{50}[μM]	Experimental IC_{50}[μM]
	4993295	2.71	>50

The compounds selected are commercially available, economically viable and passed the following criteria: predicted IC_{50}, 50μM, SVM-C scores >1.5 and overlap = 1. Compounds are tested in triplicate. Compounds with reported IC_{50} values are active across the triplicates. The reported experimental IC_{50} value is the mean of the triplicates. CID is the compound's PubChem ID number, *Compound was identified as a Factor XIIa inhibitor in our previous work.[58]

Table 5. IC_{50} of Compounds Similar to a Known PAINS at 0%, 0.01%, and 0.02% Tween20 surfactant

CID	0% surfactant	0.01% surfactant	0.02% surfactant
	IC_{50}[μM]	IC_{50}[μM]	IC_{50}[μM]
5440853	23.2	28.8	26.0

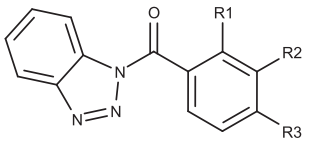
The compounds tested are aggregating without surfactant, due to changing IC_{50} at different surfactant levels. However, IC_{50} are similar between 0.01% and 0.02%, suggesting the effects of surfactants has basically plateaued at the 0.02% level of Tween20 used in the buffer.

identified by Ferreira.[78] To check for aggregation activity, the experimental validation study was repeated at different surfactant concentrations, similar to the approach from Irwin et al.[79]: 0%, 0.01%, and 0.02%. The surfactant Tween 20 was used instead of Trition-X100 used by Irwin et al., because Tween-20 was the surfactant specified by the depositors of AID 846[70] and is a nonionic surfactant-like Trition X-100. The results, shown in Table 5, suggest the 0.02% Tween 20 used in the experimental protocol was able to inhibit any aggregation that may occur and the observed activity is due to the compound and not any aggregation. Accordingly, we are confident the reported IC_{50} is due to solely to the compound and not any aggregation that may occur.

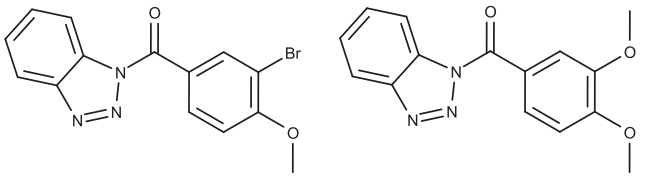
Model discussion

The previously noted increase in training and cross validation error between the first and second pass[57,58] persists in these model results as well. The previously posited explanation,[58] the difference in the mix of data available during model training between the first and second pass, seems to apply here as well. As discussed below, the experimental validation of the first pass identified one common scaffold, shown in Figure 5a, in two of the three active compounds listed in Table 1. By presenting more information on compounds containing this and other scaffolds when creating the training set for the second pass, the mix of data shifts and increasing bias for certain scaffolds and against others. This is supported by the results of the second pass experimental validation, shown in Table 4, where the active scaffold from Figure 5a is present in the majority of compounds tested. We see evidence for this effect as well in the SVM-R ROC curves when comparing Figures 3 and 4. As previously posited,[58] changing the mix and bias of the models may result in models that are less accurate (lower training and cross-validation accuracy) but more predictive (higher hit-rate).

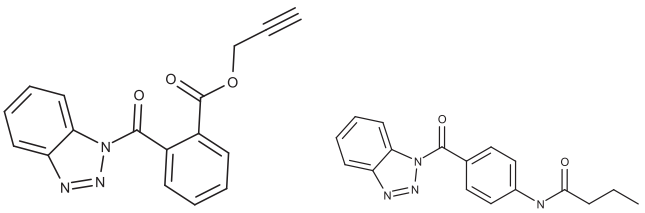
Additionally, the hit-rate of the different passes was lower in this work compared to our earlier work.[57,58] This may be due to the difference in active/inactive distributions between the previous data sets to AID 846. The previous data sets were roughly split 1:1 active: inactive while AID 846 was split 1:3



(a) The scaffold.



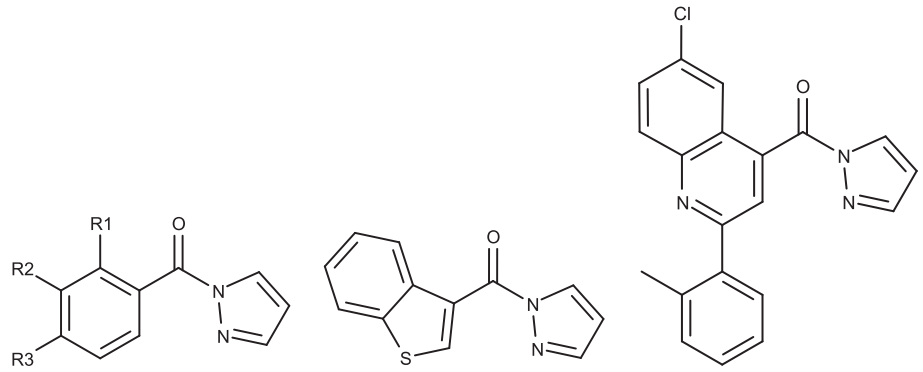
(b) Left: CID 833942, right: CID 743706.



(c) Left: CID 658532, right: CID 2201551.

Figure 5. (a) The scaffold found in the majority of the actives from the first and second pass. All but two compounds containing the scaffold in AID 846 and experimental validation work are active. Experimentally validated compounds only had functional groups in the meta and para positions (R2 and R3, respectively) while compounds in AID 846 had functional groups in all ortho, meta, and para positions(R1, R2, and R3, respectively). (b) CID 833942 is inactive while CID 743706 is. The only difference between the two is the bromine group replacing the methoxy group. Thus, the bromine group is likely preventing inhibition. (c) Both CID 658532 and 2201551 are active. However, their functional groups are much larger than others, leading to speculation there may be a degree of tolerance when it comes to the size of the functional groups.

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(a) Scaffold. (b) CID 841465. (c) CID 2220615.

Figure 6. The compounds in Table 4 containing the scaffold were inactive but both CID molecules are active. It may be inferred that, the pyrazole carbalhyde portion of the molecule does not actively contribute to inhibition.

active:inactive. Such ratios may lead to model bias against the active class though the hit-rates for our vHTS approach are still encouraging.

Factor XII discussion

In total, the experimental validations of our work have yielded 10 newly identified inhibitors for Factor XIa that may be used in future efforts to study Factor XI. While the exact mechanism by which these inhibitors work is unknown, analyzing them and other compounds for which there exists experimental data may yield inferences relating to the roles of scaffolds and functional groups.

The compounds tested can roughly be sorted into several different scaffolds. One of them, shown in Figure 5a, was present in two compounds (both active) in the first pass and six compounds (5 actives, 1 inactive) in the next pass. Examining AID 846, the scaffold was present in 8 of the 289 compounds tested. Of the eight, six were active and two were inactive. By comparing the 3 inactives to the 13 actives, several inferences can be made about this scaffold:

1. The functional groups are the main contributors to inhibition. CID 570059 from AID 846 is the base scaffold with H for all R groups. However, it is inactive. Therefore, the scaffold itself is likely responsible for providing the correct geometry and location for functional groups.
2. The bromine group may prevent Factor XIa binding. When comparing CID 833492 from AID 846 to the compound most similar to it, CID 743706 from AID 846, the only difference is the OCH_3/bromine group at the metaposition, shown in Figure 5b. Therefore, bromine is most likely preventing the molecule from binding to Factor XIa.
3. The location of the functional group on the scaffold is important. Two compounds, CID 743608 and CID 710644, from the second pass are identical except for the placement of the chlorine group: the chlorine group was in the metaposition for CID 743608 and the para position for CID 710644. However, only CID 743,608 was active. Therefore, the placement of the chlorine group on the scaffold was the determining factor of biological activity.
4. There may be a large tolerance for the size of R groups. Of the 12 active compounds, two have long/bulky R groups, CID 658532 and 2201551, shown in Figure 5c. These two have functional groups with 5 unit backbones and might be expected to inhibit binding to Factor XIa sterically. The Factor XIa may have a forgiving binding site to accommodate for the larger R groups or can encourage the R groups to fold in a way to minimize steric strain.

An active compound from the second pass, CID 5166982, seems to contain a new scaffold that is a derivative of the scaffold shown in Figure 5a. No similar structures were found in AID 846 so it may be a new scaffold for consideration. This seems to indicate the modeling and screening process is not only able to identify one scaffold from many for further analysis but also find new scaffolds for consideration. Additionally, because it is a derivative of an active scaffold, it may be able to find new scaffolds by combining the parts of different scaffolds that contribute to activity.

The other scaffold we can make inferences about in this work, shown in Figure 6, was contained in two inactive molecules. There were no compounds containing this scaffold, but two compounds that contained part of the scaffold were found and were active. By comparing and contrasting the scaffold with the two compounds, it can be inferred the pyrazole carbaldehyde portion does not actively contribute to inhibition. Because it does not change the geometry of the molecule, the pyrazole carbaldehyde may either provide no effect on inhibition or help orient it into the correct position for inhibition to occur.

Conclusions

Factor XI is a clotting factor that has important implications in external and internal clotting. Depending on different circumstances, Factor XI activation or inhibition may be desired. Different techniques to develop inhibitors have yielded some results although more is desired. As computational techniques have been shown to be beneficial in drug discovery and there are relatively few applications of it to Factor XIa inhibitor discovery, this work has applied a vHTS pipeline to find additional Factor XIa inhibitors. Screening the PubChem Compound database to focused consideration on compounds that are thermodynamically stable enough to be synthesized and cataloged in a compound library. Overall, this work has discovered 10 new inhibitors with an overall hit-rate of 45.5% (27.3% in the first pass, 63.6% in the second pass). Further analysis of the results has identified new avenues of research, including the functionality of different parts of a scaffold and

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the identification of a scaffold for further study. Ultimately, the 10 newly identified inhibitors can be used in future studies of Factor XIa, whether the mechanism or the functionality of different functional groups, or the benefits of ligand scaffolds.

Acknowledgments

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